DATABASE MANAGEMENT SYSTEM

Frontend

Focuses on layout, animations, content organization, navigation, graphics.

Programming languages:
JavaScript, HTML, CSS

Backend

Focuses on building code, debugging, database management.

Programming languages:
Node.js, Python, Java

Front End VS Back End

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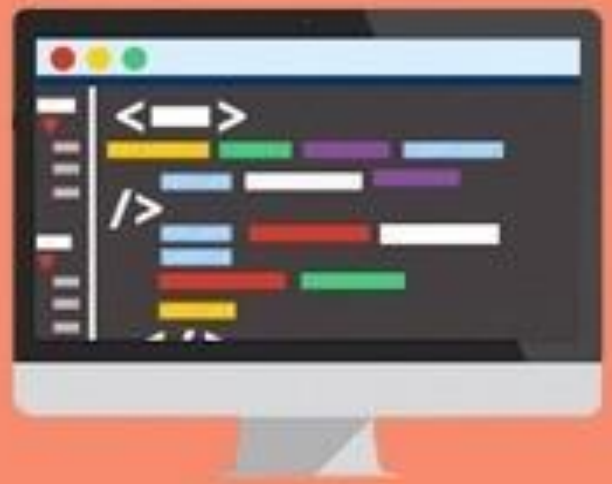
What Is The Difference?

Vs



Front End Development

Back End Development

Inbox (6) - bhabethscanonay | Meet - zyd-xfev-tfj | Facebook | front end example - Google | Online Notepad - free, no ad

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- Simple. Fast. Free. No ads. Secure - don't trust us, check the code yourself.

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★ **Click to learn** what's really going on behind the scene!

Android app

★ Sync notes
★ Lock/unlock
★ Work **offline**
★ Backup notes

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Elements | Console | Sources | Network | Performance | Memory | Application | Security | Lighthouse | Recorder | Performance insights

```
<!DOCTYPE html>
<html lang="en">
  <head> … </head>
  <body style> == $0
    <div class id="wholesite"> … </div>
    <!-- Toaster -->
    <div id="outer-toast"> … </div>
    <!-- Loader -->
    <div id="loader" class="displaynone ui-widget-overlay"> … </div>
    <script type="text/javascript"> … </script>
  </body>
</html>
```

html | body

Styles | Computed

Filter :hov

```
element.style {
}
body {
  background: ▸ #
  font: ▸ 14px/21px
    "Helvetica Ne
    Arial, sans-se
  color: #444;
  -webkit-font-smoo
    antialiased;
  -webkit-text-size
}
body {
  line height: 1;
}
html, body, div, sp
applet, object, ifr
h4, h5, h6, p, bloc
abbr, acronym, addr
code, del, dfn, em,
s, samp, small, str
sup, tt, var, b, u,
dd, ol, ul, li, fie
legend, table, capt
```

Console | What's New

Highlights from the Chrome 116 update

Improved debugging of missing stylesheets

```html
<!doctype html>
<html>
<head>
<meta charset="UTF-8">
<title>index.html</title>
</head>

<body>
    <div class="carousel">
    <div class="carousel__control">
    </div>
    <div class="carousel__stage">
        <div class="spinner spinner--left">
            <div class="spinner__face js-active" data-bg="#27323c">
                <div class="content" data-type="iceland">
                    <div class="content__left">
                        <h1>ICELAND<br><span>EUROPE</span></h1>
                    </div>
                    <div class="content__right">
                        <div class="content__main">
                            <p>"As I flew north to begin my third circuit of Iceland in four years, I was slightly anxious. The number of visitors to Iceland has doubled in that period, and I feared this might mean a little less magic to go around. At the end of this trip, 6000km later, I'm thrilled to report that the magic levels remain high. It's found in glorious football victories and Viking chants, kayaking among icebergs, sitting with puffins under the midnight sun and crunching across brand-new lava fields."
                            </p>
                          <p>- Carolyn Bain</p>
                        </div>
                        <h3 class="content__index">01</h3>
                    </div>
                </div>
            </div>
            <div class="spinner__face" data-bg="#19304a">
                <div class="content" data-type="china">
                    <div class="content__left">
                        <h1>CHINA<br><span>ASIA</span></h1>
                    </div>
                    <div class="content__right">
                        <div class="content__main">
                            <p>"Its modern face is dazzling, but China is no one-trick pony. The world's oldest continuous civilisation isn't all smoked glass and brushed aluminium and while you won't be tripping over artefacts – three decades of round-the-clock development and rash town-planning have taken their toll – rich seams of antiquity await."</p>
                          <p>- Damian Harper</p>
                        </div>
                        <h3 class="content__index">02</h3>
                    </div>
                </div>
            </div>
            <div class="spinner__face" data-bg="#2b2533">
                <div class="content" data-type="usa">
                    <div class="content__left">
                        <h1>USA<br><span>NORTH AMERICA</span></h1>
                    </div>
                    <div class="content__right">
                        <div class="content__main">
                            <p>"When it comes to travel, America has always floored me with its staggering range of possibilities. Not many other countries have so much natural beauty – mountains, beaches, rainforest, deserts, canyons, glaciers – coupled with fascinating cities to explore, an unrivaled music scene and all the things that make travel so rewarding (friendly locals, great restaurants and farmers markets, and plenty of quirky surprises)." </p>
                          <p>- Regis St Louis</p>
                        </div>
                        <h3 class="content__index">03</h3>
```

Introduction To DBMS

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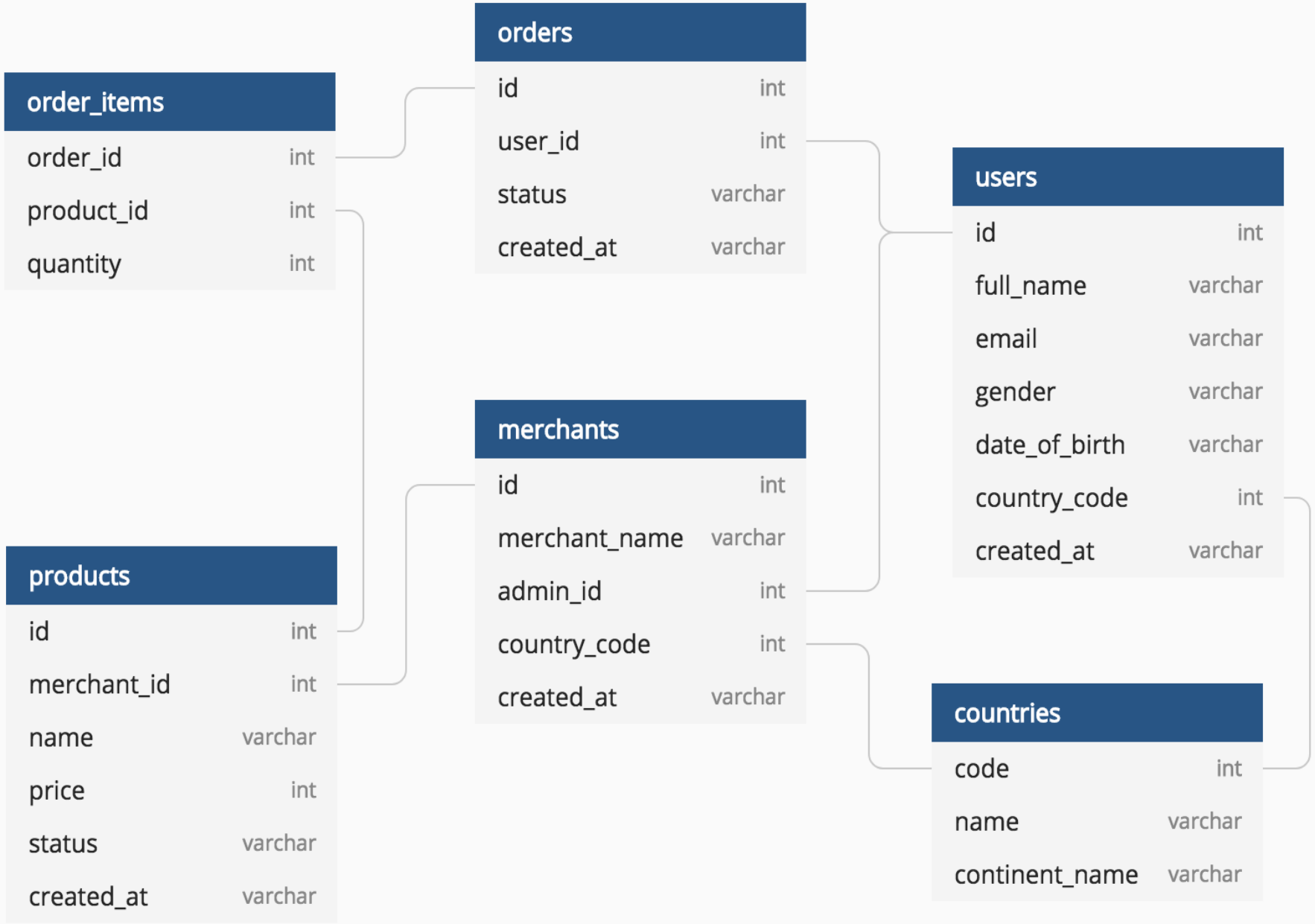
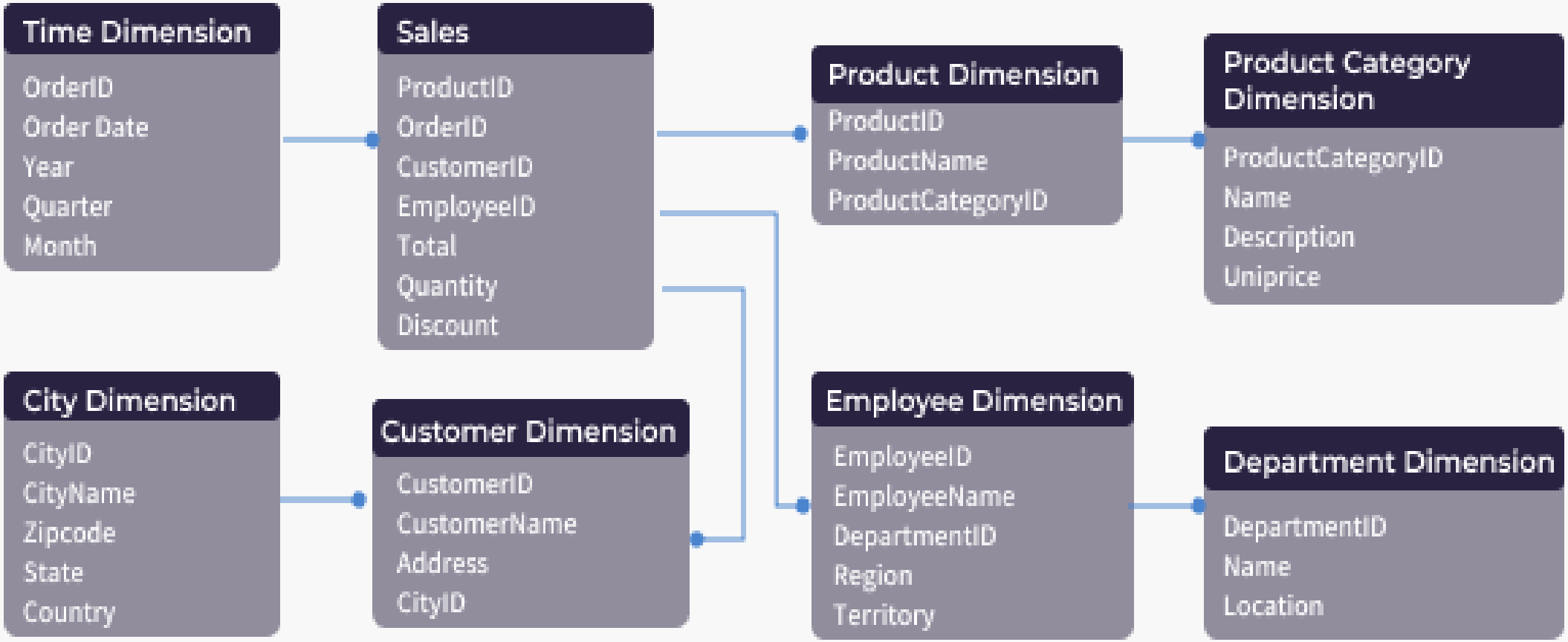

order_items
order_id int
product_id int
quantity int
orders
id int
user_id int
status varchar
created_at varchar
users
id int
full_name varchar
email varchar
gender varchar
date_of_birth varchar
country_code int
created_at varchar
merchants
id int
merchant_name varchar
admin_id int
country_code int
created_at varchar
products
id int
merchant_id int
name varchar
price int
status varchar
created_at varchar
countries
code int
name varchar
continent_name varchar

SCHEMA THEORY

Integrate.io

Database Schema Design



WHO DEVELOPED SQL PROGRAMMING LANGUAGE?

- The SQL programming language was developed in the 1970s by IBM (International Business Machine Armonk, New York) researchers Raymond Boyce and Donald Chamberlin.

IBM

ABOUT DONALD D. CHAMBERLIN

- Donald D. Chamberlin is an American computer scientist who is one of the principal designers of the original SQL language specification with Raymond Boyce. He also made significant contributions to the development of XQuery. Wikipedia
- **Born:** December 21, 1944 (age 78 years), San Jose, California, United States
- **Education:** Harvey Mudd College, Stanford University, Campbell Union High School

ABOUT RAYMOND F. BOYCE

- Raymond F. Boyce was an American computer scientist who was known for his research in relational databases. He is best known for his work co-developing the SQL database language and Boyce-Codd normal form. Wikipedia
- **Born:** 1946
- **Died:** June 16, 1974
- **Education:** Purdue University

WHAT IS DBMS?

- A Database Management System (DBMS) is software that provides an interface for users and applications to interact with databases. Its primary function is to manage and organize data, ensuring efficient storage, retrieval, manipulation, and security of the data stored within the database.

Key features and responsibilities of a DBMS include:

1. Data Storage and Retrieval: DBMS handles the physical storage of data on disk or in memory and provides mechanisms to efficiently retrieve data based on user queries.

2. Data Manipulation: It enables users to perform operations like inserting new data, updating existing data, and deleting data from the database.

3. **Data Integrity and Security:** DBMS enforces data integrity constraints (e.g., uniqueness, referential integrity) to maintain the quality and consistency of the data. It also provides access controls to ensure that only authorized users can access and modify data.

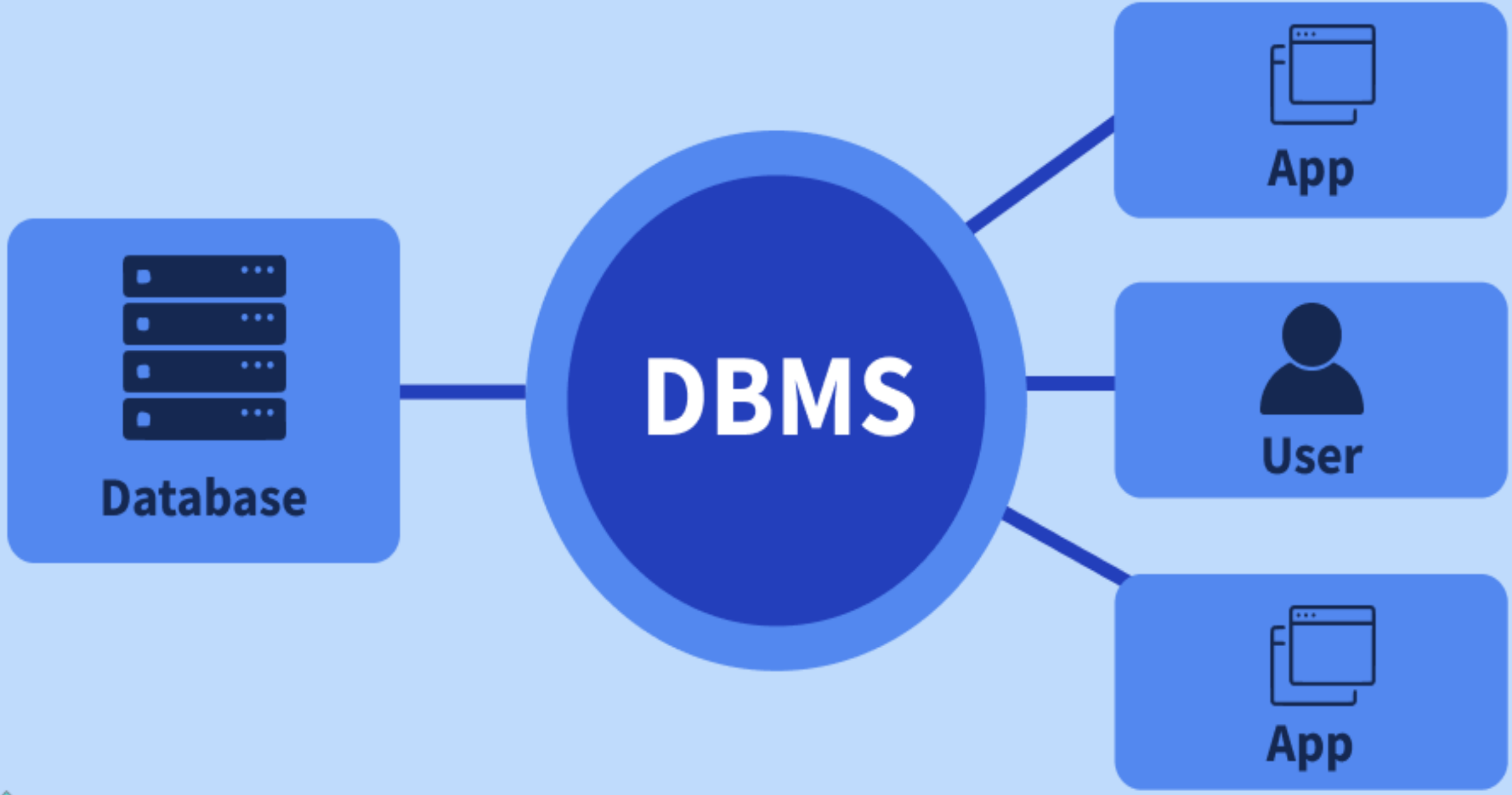
4. **Concurrency Control:** DBMS manages multiple users accessing and modifying the data simultaneously, ensuring that their actions do not interfere with each other and maintaining data consistency

5. **Query Language:** DBMS provides a standardized query language, typically SQL (Structured Query Language), which allows users to interact with the database by specifying queries to retrieve, manipulate, and manage data

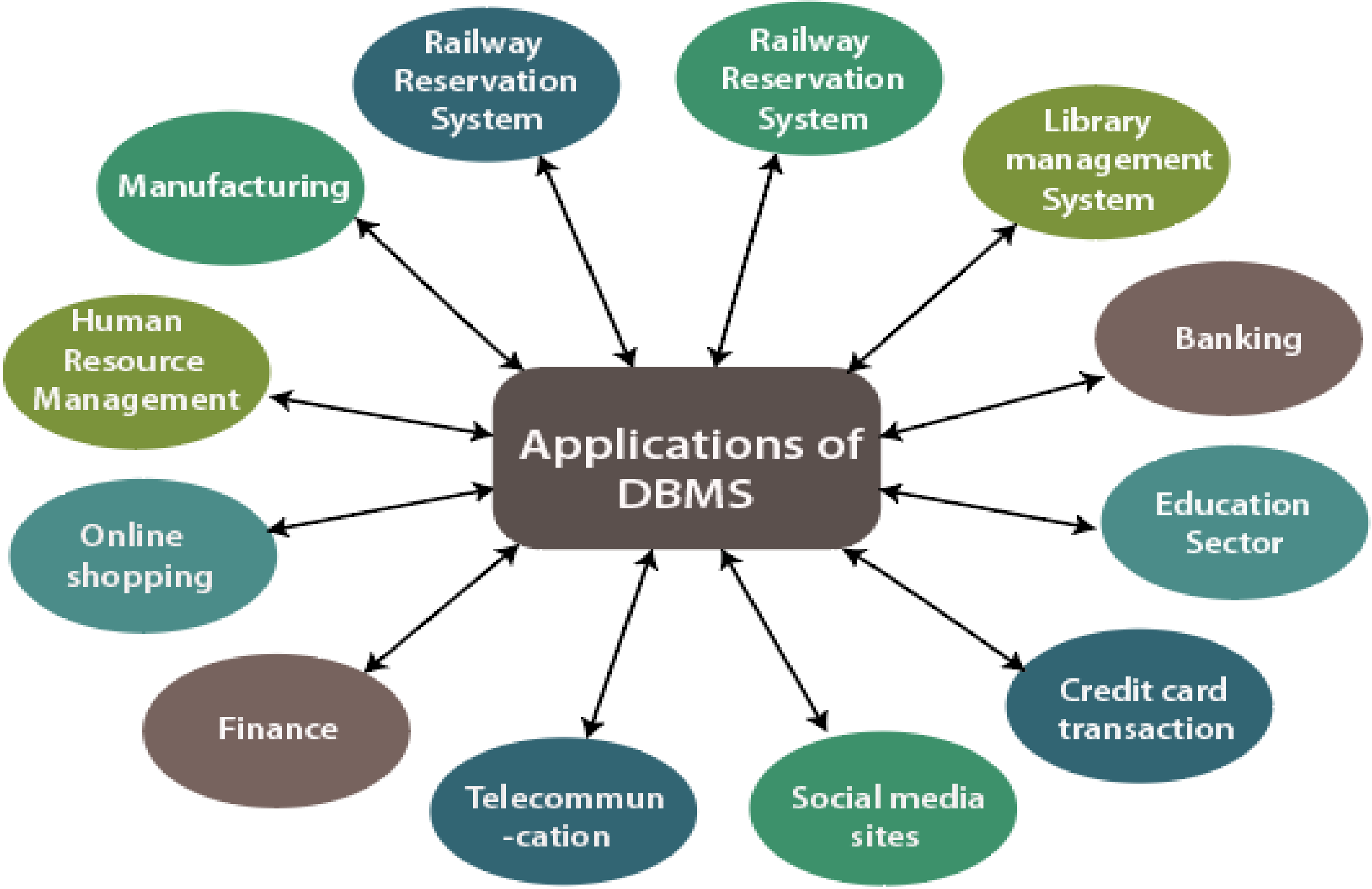
6. **Indexing and Optimization:** To improve query performance, DBMS uses indexing techniques and query optimization strategies to speed up data retrieval.

7. **Data Modeling:** DBMS often supports the creation of data models, which define the structure of the data and the relationships between different pieces of data. The most common data model for relational databases is the relational model.

8. S**calability:** Depending on the type of DBMS, it may offer scalability options to handle growing amounts of data and increasing user load.



InterviewBit


Applications of DBMS
Railway Reservation System
Railway Reservation System
Library management System
Banking
Education Sector
Credit card transaction
Social media sites
Telecommun -cation
Finance
Online shopping
Human Resource Management
Manufacturing

Applications of DBMS

DBMS - Most Popular Database Management Systems

ORACLE DATABASE

MySQL

mongoDB

Microsoft SQL Server

MariaDB

PostgreSQL

Microsoft Access

IBM DB2

THANK YOU!